



AKTU

B.TECH : First Year



Engineering Physics

UNIT - 1 : Quantum Mechanics

ONE SHOT LECTURE

- PDF NOTES (Theory + Derivation + Solved Numerical + PYQs)
- Most Important Questions
- DPP



By Gulshan Sir



AKTU B.Tech I-Year, II-Year, III-Year Gateway Classes



Courses in Application for the Session:2025-26

II-Year Courses will be available in App → Last Exam of I-Year

III-Year Courses will be available in App → Last Exam of II-Year

I-Year Courses available in App → Now Onwards



Subjects By Gulshan sir in App

S.NO	SUBJECT	SEMESTER
1	Engineering Mathematics – I	I
2	Engineering Mathematics – II	II
3	Engineering Physics	I/II
4	Fundamental of electronics engineering	I/II
5	Engineering Mathematics – III	III / IV
6	Engineering Mathematics – IV	III / IV
7	Discrete Mathematics OR Discrete Structures and Theory of Logic (DSTL)	III

UNIT - 1 : Quantum Mechanics

- Inadequacy of classical mechanics
- Planck's theory of black body radiation(qualitative)
- Compton effect
- De-Broglie concept of matter waves
- Davisson and Germer Experiment
- Phase velocity and group velocity
- Time-dependent and time-independent Schrodinger wave equations
- Physical interpretation of wave function
- Particle in a one-Dimensional box.

B.TECH : Engineering Physics

UNIT : Quantum Mechanics

Lecture-1

Today's target

- Difference between Classical Mechanics and Quantum Mechanics
- Inadequacy of classical Mechanics
- Planck's Theory of Black body radiation
- DPP
- PYQ

Classical Mechanics	Quantum Mechanics
➤ It deals with macroscopic objects ➤ It is based on Newton's laws of motion ➤ Future behavior of the particle can be identify if position of particle is known ➤ Light have wave nature ➤ Experiment such as Photoelectric effect, Compton effect, Raman effect can not be explained in classical Mechanics	➤ It deals with microscopic objects ➤ It is based on Schrodinger's wave equation ➤ Uncertainty in the determination of position and momentum of the particle ➤ Light have dual nature : Wave and Particle ➤ Experiment such as Photoelectric effect, Compton effect, Raman effect explain in Quantum Mechanics

Inadequacy of classical Mechanics

- It does not explain the behavior of microscopic particles such as atomic particles
- It could not explain a large number of observed phenomenon like
 - (1) Spectrum of Black Body Radiation
 - (2) Compton Effect
 - (3) Photoelectric Effect
 - (4) Raman Effect etc.
- It could not explain the stability of atoms
- It could not explain the observed variation of specific heat of metal and gas

Note : The inadequacy of classical mechanics led to the development of Quantum Mechanics.

- A perfect Black Body is one which can absorb all the radiations of all wavelength incident on it.
- When a black body is heated, it emits all types of radiations of all wavelength called black body radiations
- A Black Body is a good absorber as well as good emitter of all radiation of all wavelength.
- Black Body are of two types
 - (1) **Natural black body** : Lamp black, Platinum black, Sun, Stars etc.
 - (2) **Artificial black body** : Electric heater, Ferry's black body
- A perfect black body can not be realized in practice but materials like Lamp black and Platinum black come close to perfect black body
- Nearest example of perfect black body is Ferry's black body

Note :

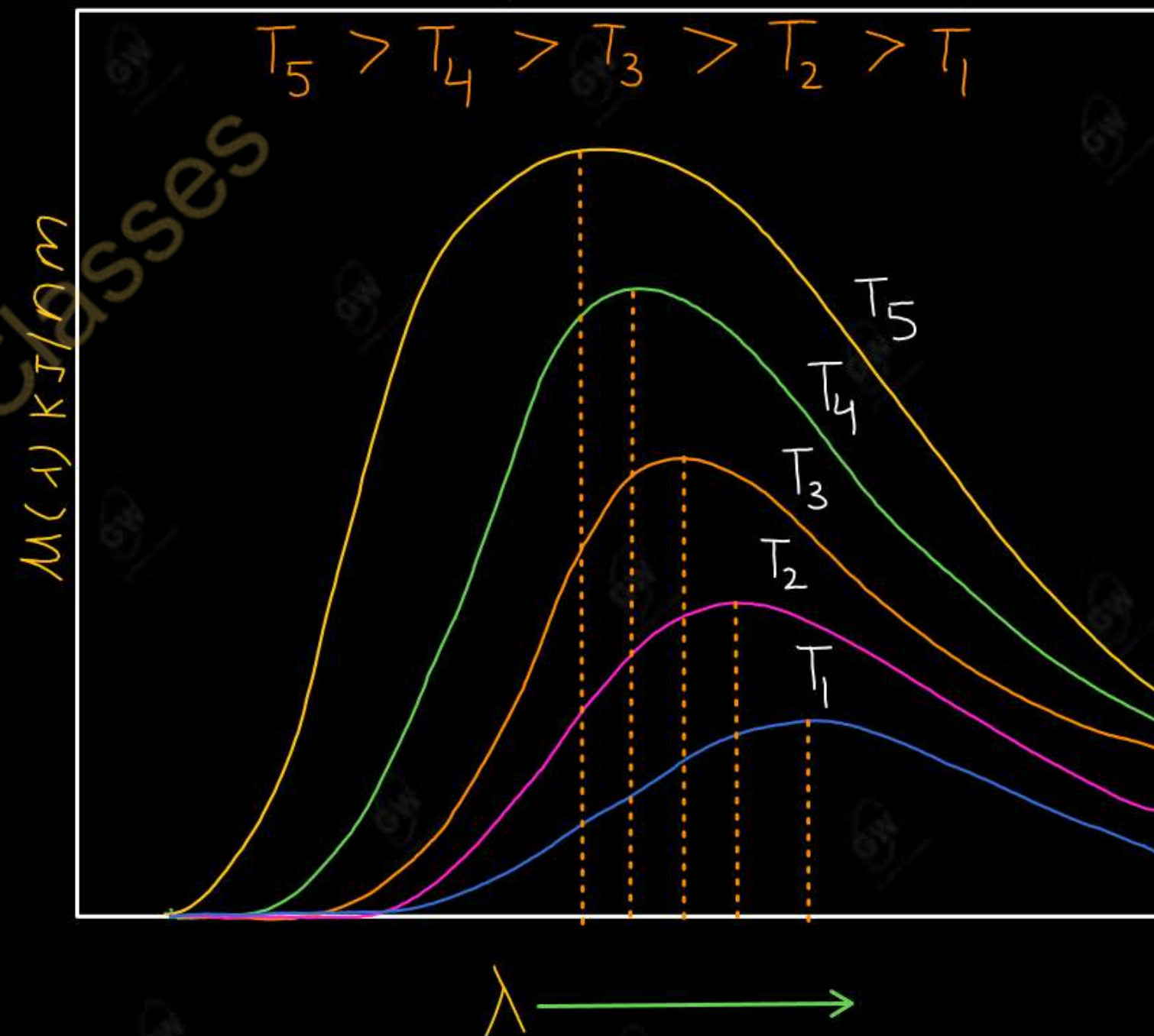
Black colored object : A body which absorb all the radiations present in visible light or white light

Energy Distribution in the spectrum of Black Body Radiation

- The energy distribution in the spectrum of black body radiation at different temperatures is shown by the graph between λ and E_λ

From the graph following conclusion can be drawn

- The intensity of radiation increases with wavelength and reaches a maximum value at a particular wavelength and decreases further as wavelength increases
- Wavelength corresponding to maximum intensity is $\lambda_{\max}$
- At each temperature of black body, there is a wavelength ($\lambda_{\max}$) for which E_λ is max.
- As temperature of body increases, the $\lambda_{\max}$ shifts towards shorter wavelength



- There is non-uniform distribution of energy in the spectrum of black body radiation at a given temperature.
- The amount of energy emitted per second increases with increase in temperature of black body.
- The area under each curve represents the total energy emitted by black body at given temperature.

Stefan's law

- According to Stefan's law, the total amount of radiant energy by a black body per unit area per unit time due to all the wavelength is directly proportional to the fourth power of absolute temperature.

$$E \propto T^4$$

$$E = \sigma T^4$$

Where σ is called the 'Stefan constant'

$$\sigma = 5.56704 \times 10^{-8} \text{ W/m}^2\text{K}^4$$

Wien's Displacement law OR Wien's first law

- According to Wien's Displacement law, λ_m is inversely proportional to the absolute temperature (T) of the body

$$\lambda_m \propto \frac{1}{T}$$

$$\lambda_m = \frac{\text{constant}}{T}$$

$$\lambda_m \times T = \text{constant}$$

Wien's Radiation formula OR Wien's second law

- According to Wien's law, the amount of radiant energy emitted by a black body per unit area per unit time for a given wavelength range λ and $\lambda + d\lambda$ at a given temperature is expressed as

$$E_{\lambda} d\lambda = \frac{8\pi hc}{\lambda^5} \left(\frac{d\lambda}{e^{\frac{hc}{\lambda KT}}} \right)$$

Where $h = 6.63 \times 10^{-34} \text{ J} - \text{sec}$

$$c = 3 \times 10^8 \text{ m/Sec}$$

$$k \text{ (Boltzmann's constant)} = 1.38 \times 10^{-23} \text{ J/K}$$

Limitation : This law is applicable for shorter wavelength only.

Rayleigh – Jeans Law

- Rayleigh used the classical theories of electro magnetism and thermodynamics to show that the black body spectral energy is given by

$$E_{\lambda} d\lambda = \frac{8\pi kT}{\lambda^4} d\lambda$$

Limitation : This law is applicable for longer wavelength only.

Engineering Physics (by Gulshan Sir)

UNIT - 1 : DPP-1

Quantum Mechanics

Q.1 What is Stefan's law ?

AKTU 2015

Q.2 What is Wein's law ?

AKTU 2018

Q.3 State Wein's Displacement law and Rayleigh-Jeans law.

AKTU 2021

Q.4 Describe energy distribution in black body radiation.

AKTU 2022

UNIT : Quantum Mechanics

Lecture-2

Today's target

- Planck's Quantum Hypothesis
- Planck's Quantum Hypothesis
- Derivation of Planck's Radiation formula
- AKTU PYQs

- In 1900 Max Planck presented a paper to explain Black body radiation.
- According to Max Planck radiation behave like particles and consist of tiny packet of energy.
- Each packet of energy is called photon or Quanta and have energy.

$$E = h\nu$$

- Max Planck proposed that the correct result can be obtained if the emission and absorption of energy is not taken as continuous process but it takes place in discrete amount.
- He derived radiation formula by using following assumptions

Basic Assumptions of Planck's hypothesis

- He assumed that the atoms of the wall of a black body behave as oscillators and each oscillator has characteristic frequency of oscillation.
- The oscillators emit or absorb energy only in the form packets of energy $h\nu$ while jumping from one Quantum state to other.

The energy of an oscillator can have only discrete values

$$E_n = nh\nu$$

$$E_1 = h\nu$$

$$E_2 = 2h\nu$$

$$E_3 = 3h\nu$$

and so on

Where

n = positive integer / Quantum Number.

h = Planck's constant

ν = Frequency of oscillation.

Planck's Radiation Formula / Planck's Radiation Law

Max Planck developed a theory of black body radiation that leads to an equation for the energy density of the radiation. This equation is in complete agreement with experimental observations.

Derivation of Planck's Radiation Formula

Let N = Total number of Oscillators

E = Total Energy of Oscillators

E_{avg} = Average Energy of Oscillators

$$E_{avg} = \frac{E}{N} \quad \text{--- (1)}$$

Let $N_0, N_1, N_2, \dots, N_n, \dots$ be the number of oscillators having energies $0, h\nu, 2h\nu, 3h\nu, \dots, nh\nu, \dots$ respectively

According to Boltmann's Law, the number of oscillators having $nh\nu$ energy at

temperature T will be

$$N_n = N_0 e^{-\frac{nh\nu}{kT}}$$

$$N_0 = N_0$$

$$N_1 = N_0 e^{-\frac{h\nu}{kT}}$$

$$N_2 = N_0 e^{-\frac{2h\nu}{kT}}$$

$$N_3 = N_0 e^{-\frac{3h\nu}{kT}}$$

and so on

Total Number of Oscillators (N)

$$N = N_0 + N_1 + N_2 + N_3 + \dots$$

$$N = N_0 + N_0 e^{-\frac{h\nu}{RT}} + N_0 e^{-\frac{2h\nu}{RT}} + N_0 e^{-\frac{3h\nu}{RT}} + \dots$$

$$N = N_0 \left(1 + e^{-\frac{h\nu}{RT}} + e^{-\frac{2h\nu}{RT}} + e^{-\frac{3h\nu}{RT}} + \dots \right)$$

$$N = N_0 \left[1 + e^{-\frac{h\nu}{RT}} + \left(e^{-\frac{h\nu}{RT}} \right)^2 + \left(e^{-\frac{h\nu}{RT}} \right)^3 + \dots \right]$$

Using

$$(1-x)^{-1} = 1 + x + x^2 + x^3 + \dots$$

$$N = N_0 \left(1 - e^{-\frac{h\nu}{RT}} \right)^{-1}$$

$$N = \frac{N_0}{\left(1 - e^{-\frac{h\nu}{RT}} \right)} \quad \text{--- (2)}$$

Total Energy (E)

$$E = N_0 \times 0 + N_1 \times h\nu + N_2 \times 2h\nu +$$

$$N_3 \times 3h\nu + \dots$$

$$E = N_1 h\nu + 2N_2 h\nu + 3N_3 h\nu + \dots$$

$$E = h\nu [N_1 + 2N_2 + 3N_3 + \dots]$$

$$E = h\nu \left[N_0 e^{-\frac{h\nu}{RT}} + 2N_0 e^{-\frac{2h\nu}{RT}} + 3N_0 e^{-\frac{3h\nu}{RT}} + \dots \right]$$

$$E = h\nu N_0 e^{-\frac{h\nu}{RT}} \left[1 + 2e^{-\frac{h\nu}{RT}} + 3e^{-\frac{2h\nu}{RT}} + \dots \right]$$

$$E = h\nu N_0 e^{-\frac{h\nu}{RT}} \left[1 + 2e^{-\frac{h\nu}{RT}} + 3\left(e^{-\frac{h\nu}{RT}}\right)^2 + \dots \right]$$

Using

$$(1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \dots$$

$$E = h\nu N_0 e^{-\frac{h\nu}{RT}} \left(1 - e^{-\frac{h\nu}{RT}} \right)^{-2}$$

$$E = \frac{h\nu N_0 e^{-\frac{h\nu}{RT}}}{\left(1 - e^{-\frac{h\nu}{RT}} \right)^2} \quad \text{--- (3)}$$

Put E and N in eqⁿ ①

$$E_{avg} = \frac{h\nu N_0 e^{-\frac{h\nu}{RT}}}{\left(1 - e^{-\frac{h\nu}{RT}}\right)^2} \times \frac{\left(1 - e^{-\frac{h\nu}{RT}}\right)}{N_0}$$

$$E_{avg} = \frac{h\nu}{e^{\frac{h\nu}{RT}} \left(1 - e^{-\frac{h\nu}{RT}}\right)}$$

$$E_{avg} = \frac{h\nu}{\left(e^{\frac{h\nu}{RT}} - 1\right)}$$

The number of oscillators per unit volume lying in the frequency range of ν to $\nu + d\nu$ as obtain by Rayleigh and Jeans

$$= \frac{8\pi\nu d\nu}{c^3}$$

Energy of radiation per unit volume

OR

Energy density ($u_\nu d\nu$)

$$u_\nu d\nu = \frac{8\pi h \nu^3 d\nu}{c^3 \left(e^{\frac{h\nu}{kT}} - 1 \right)}$$

Planck's Radiation Formula in terms of frequency (ν)

Put

$$\nu = \frac{c}{\lambda}$$

$$d\nu = -\frac{c}{\lambda^2} d\lambda$$

$$u_\lambda d\lambda = -u_\nu d\nu$$

$$u_\lambda d\lambda = \frac{8\pi h \cancel{c^3} c d\lambda}{\cancel{c^3} \lambda^3 \lambda^2 \left(e^{\frac{hc}{\lambda kT}} - 1 \right)}$$

$$u_\lambda d\lambda = \frac{8\pi h c d\lambda}{\lambda^5 \left(e^{\frac{hc}{\lambda kT}} - 1 \right)}$$

Planck's radiation formula in terms of Wavelength (λ)

Deduction of Various laws using Planck's Radiation Formula

(i) Wien's law from Planck's Radiation Formula

$$u_{\lambda} d\lambda = \frac{8\pi hc}{\lambda^5} \frac{d\lambda}{\left(e^{\frac{hc}{\lambda KT}} - 1\right)}$$

For short wavelength or When λ is small

$$e^{\frac{hc}{\lambda KT}} \gg 1$$

So, 1 can be neglected

$$u_{\lambda} d\lambda = \frac{8\pi hc}{\lambda^5} \frac{d\lambda}{e^{\frac{hc}{\lambda KT}}}$$

This is the Wien's law agree with experimental curve at short wavelengths region.

(ii) Rayleigh – Jeans law from Planck's Radiation Formula

$$u_{\lambda} d\lambda = \frac{8\pi hc}{\lambda^5} \frac{d\lambda}{\left(e^{\frac{hc}{\lambda KT}} - 1\right)}$$

For longer wavelength or When λ is large

$$e^{\frac{hc}{\lambda KT}} = 1 + \frac{hc}{\lambda KT}$$

$$u_{\lambda} d\lambda = \frac{8\pi hc}{\lambda^5} \frac{d\lambda}{\left(1 + \frac{hc}{\lambda KT} - 1\right)}$$

$$u_{\lambda} d\lambda = \frac{8\pi KT}{\lambda^4} d\lambda$$

This is the Rayleigh – Jeans law agree with experimental curve at long wavelengths region

Q.1 Calculate the energy of oscillator of frequency 5.3×10^{12} /sec at 500 K treating it as

(i) Classical oscillator

(ii) Planck's oscillator.

Given that $k = 1.38 \times 10^{-23}$ J/K and $h = 6.6 \times 10^{-34}$ J – sec

AKTU-2023-24

Given

$$\nu = 5.3 \times 10^{12} \text{ Hz}$$

$$T = 500 \text{ K}$$

$$k = 1.38 \times 10^{-23} \text{ J/K}$$

$$h = 6.6 \times 10^{-34} \text{ J-sec}$$

(i) Classical oscillator

$$E_{avg} = kT$$

$$= 1.38 \times 10^{-23} \times 500$$

$$E_{avg} = 6.9 \times 10^{-21} \text{ J}$$

(ii) Planck's oscillator

$$E_{avg} = \frac{h\nu}{e^{\frac{h\nu}{kT}} - 1}$$

$$h\nu = 6.6 \times 10^{-34} \times 5.3 \times 10^{12}$$

$$h\nu = 3.498 \times 10^{-21} \text{ J}$$

$$\frac{h\nu}{kT} = \frac{6.6 \times 10^{-34} \times 5.3 \times 10^{12}}{1.38 \times 10^{-23} \times 500}$$

$$= 0.507$$

$$E_{avg} = \frac{3.498 \times 10^{-21}}{e^{0.507} - 1}$$

$$E_{avg} = 5.3 \times 10^{-21} \text{ J}$$

Gateway Classes

UNIT-1 : DPP-2**Quantum Mechanics**

Q.1 Write down the Basic Assumptions of Planck's hypothesis?

AKTU-2018-19

Q.2 Derive the Planck's Radiation Formula.

AKTU-2016-17

Q.3 Calculate the energy of oscillator of frequency 4.2×10^{12} Hz at 27°C treating it as

(i) classical oscillator

(ii) Planck's oscillator

Given that $k = 1.38 \times 10^{-23} \text{ J/K}$ and $h = 6.63 \times 10^{-34} \text{ J} \cdot \text{sec}$

AKTU-2018-19

UNIT : Quantum Mechanics

Lecture - 3

Today's target

- Compton Effect
- DPP
- PYQs

Gateway Classes

- In 1921, Professor A.H Compton discovered that when a monochromatic beam of high frequency radiation (X – ray) is scattered by a substance (or electron) then the scattered radiations contain the radiation not only of the same wavelength or frequency as that of incident ray but also the radiation of greater wavelength or lower frequency
- The radiation of unchanged wavelength in the scattered rays are called Unmodified radiation
- The radiation of greater wavelength in the scattered rays are called modified radiation
- The phenomenon of change in wavelength of scattered X-ray is known as **Compton effect**.
- The change in wavelength of high frequency radiation during scattering process is called **Compton shift**.

$$\Delta\lambda = \lambda' - \lambda$$

Where $\Delta\lambda$ = Compton Shift

λ' = Wavelength of scattered radiation

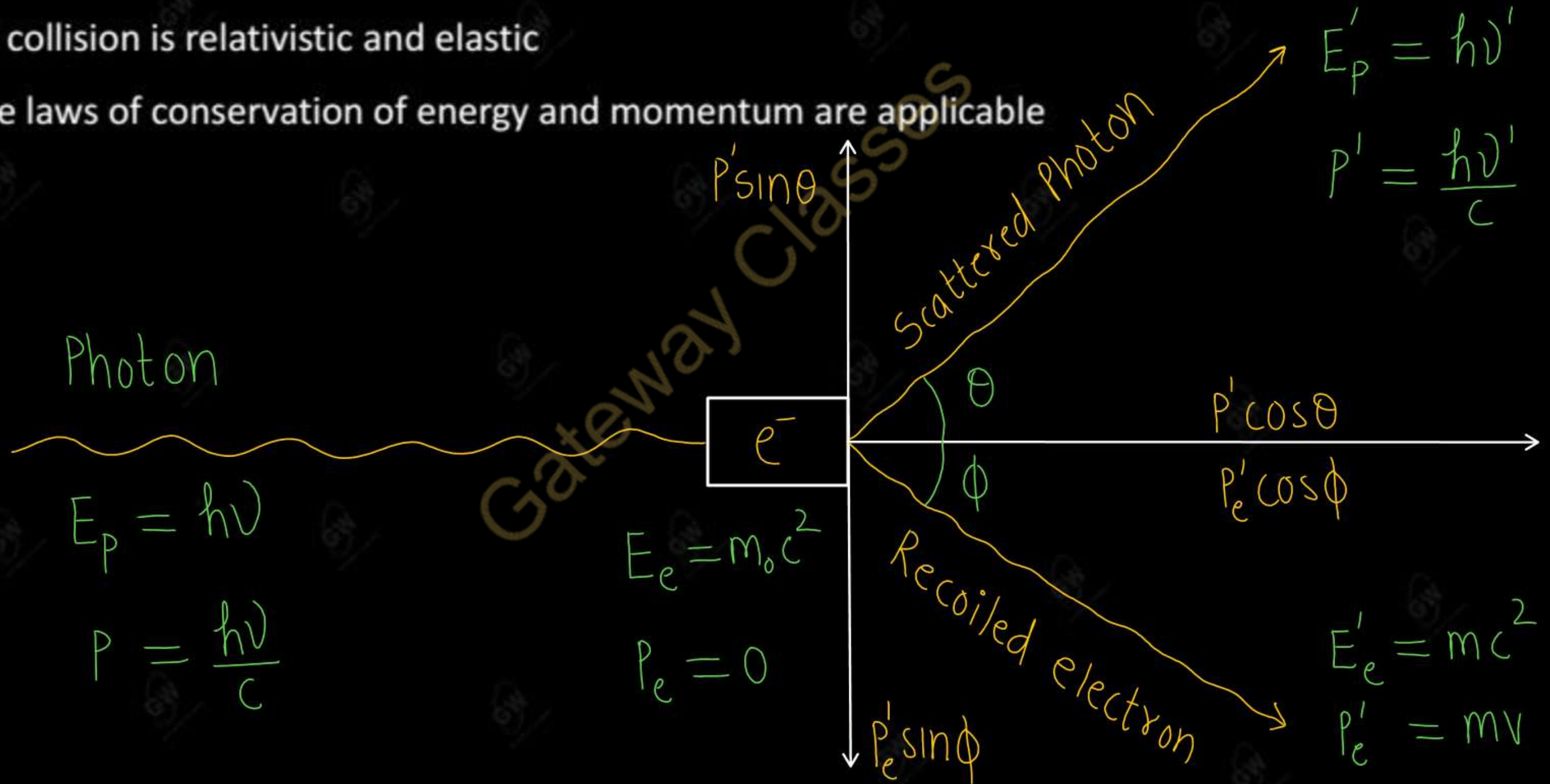
λ = Wavelength of incident radiation

➤ Following assumption were made to derive the Compton shift

(i) Compton effect is the result of interaction of X-ray photon and free electron of the target.

(ii) The collision is relativistic and elastic

(iii) The laws of conservation of energy and momentum are applicable



Apply Law of conservation of momentum

Momentum Before collision = Momentum After collision

Along X-axis

$$P + P_e = P' \cos \theta + P_e' \cos \phi$$

$$\frac{h\nu}{c} + 0 = \frac{h\nu'}{c} \cos \theta + mv \cos \phi$$

$$\frac{h\nu}{c} = \frac{h\nu' \cos \theta + mc v \cos \phi}{c}$$

$$h\nu = h\nu' \cos \theta + mc v \cos \phi$$

$$mc v \cos \phi = h\nu - h\nu' \cos \theta$$

①

Along Y-axis

$$0 + 0 = P' \sin \theta - P_e' \sin \phi$$

$$0 = \frac{h\nu'}{c} \sin \theta - mv \sin \phi$$

$$0 = \frac{h\nu' \sin \theta - mc v \sin \phi}{c}$$

$$0 = h\nu' \sin \theta - mc v \sin \phi$$

$$mc v \sin \phi = h\nu' \sin \theta$$

②

Squaring and Adding eqⁿ ① and ②

$$(mcv \cos \phi)^2 + (mcv \sin \phi)^2 = (h\nu - h\nu' \cos \theta)^2 + (h\nu' \sin \theta)^2$$

$$m^2 c^2 v^2 \cos^2 \phi + m^2 c^2 v^2 \sin^2 \phi = h^2 \nu^2 + h^2 \nu'^2 \cos^2 \theta - 2 h^2 \nu \nu' \cos \theta + h^2 \nu'^2 \sin^2 \theta$$

$$m^2 c^2 v^2 (\sin^2 \phi + \cos^2 \phi) = h^2 \nu^2 + h^2 \nu'^2 (\sin^2 \theta + \cos^2 \theta) - 2 h^2 \nu \nu' \cos \theta$$

$$m^2 c^2 v^2 = h^2 \nu^2 + h^2 \nu'^2 - 2 h^2 \nu \nu' \cos \theta \quad \text{————— ③}$$

Apply Law of conservation of energy

Energy Before collision = Energy After collision

$$E_p + E_e = E'_p + E'_e$$

$$h\nu + m_0 c^2 = h\nu' + mc^2$$

$$mc^2 = (h\nu - h\nu') + m_0 c^2$$

Squaring both side

$$m^2 c^4 = [(h\nu - h\nu') + m_0 c^2]^2$$

$$m^2 c^4 = (h\nu - h\nu')^2 + m_0^2 c^4 + 2(h\nu - h\nu') m_0 c^2$$

$$m^2 c^4 = h^2 \nu^2 + h^2 \nu'^2 - 2h^2 \nu \nu' + m_0^2 c^4 + 2(h\nu - h\nu') m_0 c^2 \quad \text{--- (4)}$$

Subtract eqⁿ (3) from (4)

$$m^2 c^4 - m^2 c^2 v^2 = h^2 \nu^2 + h^2 \nu'^2 - 2h^2 \nu \nu' + m_0^2 c^4 + 2(h\nu - h\nu') m_0 c^2 - h^2 \nu^2 - h^2 \nu'^2 + 2h^2 \nu \nu' \cos \theta$$

$$m^2 c^4 - m^2 c^2 v^2 = -2h^2 \nu \nu' + m_0^2 c^4 + 2(h\nu - h\nu') m_0 c^2 + 2h^2 \nu \nu' \cos \theta$$

$$m^2 c^4 \left(1 - \frac{v^2}{c^2}\right) = -2h^2 \nu \nu' (1 - \cos \theta) + m_0 c^4 + 2(h\nu - h\nu') m_0 c^2 \quad \text{--- (5)}$$

According to theory of Relativity Using eqⁿ (6) in (5)

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$m^2 = \frac{m_0^2}{\left(1 - \frac{v^2}{c^2}\right)}$$

$$m^2 \left(1 - \frac{v^2}{c^2}\right) = m_0^2 \quad \text{--- (6)}$$

$$\cancel{m_0 c^4} = -2h^2 \nu \nu' (1 - \cos \theta) + \cancel{m_0 c^4} + 2(h\nu - h\nu') m_0 c^2$$

$$2h^2 \nu \nu' (1 - \cos \theta) = 2h(\nu - \nu') m_0 c^2$$

$$h\nu \nu' (1 - \cos \theta) = (\nu - \nu') m_0 c^2$$

$$\frac{h(1 - \cos \theta)}{m_0 c^2} = \frac{\nu - \nu'}{\nu \nu'}$$

$$\frac{h(1 - \cos \theta)}{m_0 c^2} = \frac{1}{\nu'} - \frac{1}{\nu}$$

$$\frac{h(1-\cos\theta)}{m_0 c} = \frac{c}{\nu'} - \frac{c}{\nu}$$

$$\frac{h(1-\cos\theta)}{m_0 c} = \lambda' - \lambda$$

$$\lambda - \lambda' = \frac{h}{m_0 c} (1 - \cos\theta)$$

$$\Delta \lambda = \frac{h}{m_0 c} (1 - \cos\theta)$$

Kinetic Energy of Recoiled electron

$$E_K = h\nu - h\nu'$$

$$= h(\nu - \nu')$$

$$= h\left(\frac{c}{\lambda} - \frac{c}{\lambda'}\right)$$

$$= hc\left(\frac{1}{\lambda} - \frac{1}{\lambda'}\right)$$

$$E_K = hc\left(\frac{\lambda' - \lambda}{\lambda \lambda'}\right)$$

CASE - 1

When $\theta = 0^\circ$

$$\Delta\lambda = \frac{h}{m_0 c} (1 - \cos\theta)$$

$$\Delta\lambda = \frac{h}{m_0 c} (1 - \cos 0^\circ)$$

$$\Delta\lambda = 0$$

CASE - 2

When $\theta = 90^\circ$

$$\Delta\lambda = \frac{h}{m_0 c} (1 - \cos\theta)$$

$$\Delta\lambda = \frac{h}{m_0 c} (1 - \cos 90^\circ)$$

$$\Delta\lambda = \frac{h}{m_0 c}$$

$$\Delta\lambda = 0.0243 \text{ \AA}$$

CASE - 3

When $\theta = 180^\circ$

$$\Delta\lambda = \frac{h}{m_0 c} (1 - \cos\theta)$$

$$\Delta\lambda = \frac{h}{m_0 c} (1 + 1)$$

$$\Delta\lambda = \frac{2h}{m_0 c}$$

$$\Delta\lambda = 0.0486 \text{ \AA}$$

Conclusion from Compton scattering Experiment

- Compton shift is given by

$$\Delta\lambda = \lambda' - \lambda = \frac{h}{m_0c}(1 - \cos\theta)$$

- It is concluded from above equation that

- (i) Wavelength of scattered photon is greater than the wavelength of incident photon.
- (ii) Compton shift only depends on the scattering angle.
- (iii) Compton shift have the same value for all the substance containing free electrons.

Why Compton effect is not observed in visible spectrum ?

The maximum change in wavelength $\lambda_{\max}$ is Approx. $0.05A^0$. This is very small therefore can not be observed for wavelength longer than few angstrom units.

Example

- (i) For X-ray, wavelength of incident radiation is about $1A^0$

$$\lambda_{\max} = 0.05A^0$$

Which is 5% (detectable) of incident radiation

- (ii) For visible radiation, wavelength of incident radiation is about $5000A^0$

$$\lambda_{\max} = 0.05A^0$$

Which is 0.001% (not detectable) of incident radiation

Q.1 In a Compton scattering experiment X-Ray of wavelength $0.015\text{\AA}$ is scattered at 60° , find the wavelength of scattered X-ray.

Given

$$\lambda = 0.015 \text{\AA}$$

$$\theta = 60^\circ$$

$$h = 6.63 \times 10^{-34} \text{ J-sec}$$

$$m_0 = 9.1 \times 10^{-31} \text{ kg}$$

$$c = 3 \times 10^8 \text{ m/s}$$

$$\lambda' = ?$$

$$\Delta \lambda = \frac{h}{m_0 c} (1 - \cos \theta)$$

$$\Delta \lambda = \frac{6.63 \times 10^{-34}}{9.1 \times 10^{-31} \times 3 \times 10^8} (1 - \cos 60)$$

$$\lambda' - \lambda = 0.012 \times 10^{-10} \text{ m} = 0.012 \text{\AA}$$

$$\lambda' = 0.015 + 0.012$$

$$\lambda' = 0.027 \text{\AA}$$

In An X- ray photon is found to have its wavelength doubled on being scattered through 90° from a material. Find the wavelength and energy of incident photon.

AKTU 2022-23

Given

$$\lambda' = 2\lambda$$

$$\theta = 90^\circ$$

$$h = 6.63 \times 10^{-34} \text{ J-sec}$$

$$m_0 = 9.1 \times 10^{-31} \text{ kg}$$

$$c = 3 \times 10^8 \text{ m/s}$$

$$\lambda = ?$$

$$E = ?$$

$$\Delta \lambda = \frac{h}{m_0 c} (1 - \cos \theta)$$

$$\lambda' - \lambda = \frac{6.63 \times 10^{-34}}{9.1 \times 10^{-31} \times 3 \times 10^8} (1 - \cos 90^\circ)$$

$$2\lambda - \lambda = 0.0245 \text{ \AA}$$

$$\lambda = 0.0245 \text{ \AA}$$

$$E = h\nu$$

$$E = \frac{hc}{\lambda}$$

$$E = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{0.0245 \times 10^{-10}}$$

$$E = 8.106 \times 10^{-14} \text{ J}$$

Q.3 X-ray of wavelength $2 \text{ \AA}$ are scattered from a black body and x-ray are scattered at angle of 45° . Calculate the Compton shift, wavelength of scattered photon.

AKTU 2018-19

Given

$$\lambda = 2 \text{ \AA}$$

$$\theta = 45^\circ$$

$$h = 6.63 \times 10^{-34} \text{ J-sec}$$

$$c = 3 \times 10^8 \text{ m/s}$$

$$m_0 = 9.1 \times 10^{-31} \text{ kg}$$

$$\Delta \lambda = ?$$

$$\lambda' = ?$$

$$\Delta \lambda = \frac{h}{m_0 c} (1 - \cos \theta)$$

$$\Delta \lambda = \frac{6.63 \times 10^{-34}}{9.1 \times 10^{-31} \times 3 \times 10^8} (1 - \cos 45^\circ)$$

$$\Delta \lambda = 0.007 \text{ \AA}$$

$$\lambda' - \lambda = 0.007 \text{ \AA}$$

$$\lambda' = 2 + 0.007 \text{ \AA}$$

$$\lambda' = 2.007 \text{ \AA}$$

UNIT – 1 : DPP-3

Quantum Mechanics

Q.1 Explain modified and unmodified radiation in Compton effect

AKTU 2016-17

Q.2 What is Compton effect and Compton shift ? Derive the expression for Compton shift

AKTU 2016-17, 2017-18, 2018-19, 2019-20, 2020-21, 2021-22

Q.3 Calculate Compton shift if x-ray of wavelength of $1.00 \text{ \AA}$ are scattered from a carbon block.

The scattered radiation is viewed at 90° to the incident

AKTU 2016-17

Ans $0.02357 \text{ \AA}$

B.TECH : Engineering Physics

UNIT : Quantum Mechanics

Lecture-4

Today's target

- De-Broglie concept of matter waves
- DPP
- PYQ

1924

Light

Wave

Particle

Gateway Classes

- According to de-Broglie Hypothesis (1924), "Every moving particle has a wave associated with it"
- The wave associated with a moving particle is called matter waves or de Broglie waves.
- If a particle of mass 'm' has momentum 'p' and λ is the wavelength of wave associated with it, then according to de-Broglie Hypothesis.

$$\lambda = \frac{h}{p}$$

Planck's constant

de-broglie wavelength

$$p = mv$$

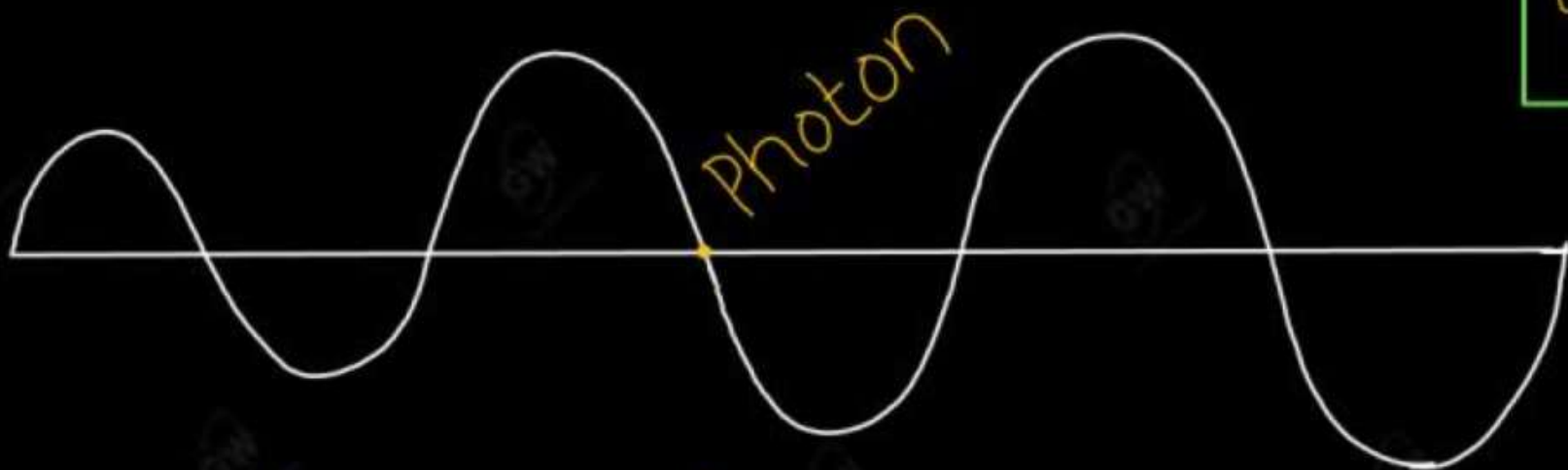
m = Mass of particle

v = velocity of particle

$$\lambda = \frac{h}{mv}$$

Expression for de-Broglie wavelength of photon

$$\nu = \frac{c}{\lambda}$$



According to Planck's theory of Radiation

$$E = h\nu \quad \text{--- (1)}$$

According to Einstein Mass energy relation

$$E = mc^2 \quad \text{--- (2)}$$

Where

E = energy of photon

From (1) and (2)

$$h\nu = mc^2$$

$$\frac{hc}{\lambda} = mc^2$$

$$\lambda = \frac{h}{mc} \quad \checkmark \checkmark$$

Some other important Expression for De-Broglie wavelength

1. In terms of kinetic energy

We know that

$$E = \frac{1}{2} m v^2$$

$$E = \frac{1}{2} \frac{m^2 v^2}{m}$$

$$E = \frac{1}{2m} (mv)^2$$

$$E = \frac{1}{2m} p^2$$

$$p^2 = 2mE$$

$$p = \sqrt{2mE}$$

de-broglie wavelength

$$\lambda = \frac{h}{p}$$

$$\lambda = \frac{h}{\sqrt{2mE}}$$



2. In terms of Temperature

We know that

$$E = \frac{3}{2} kT$$

k = Boltzmann's constant

T = Temperature

E = kinetic energy

de-broglie wavelength

$$\lambda = \frac{h}{\sqrt{2mE}}$$

$$\lambda = \frac{h}{\sqrt{2m \times \frac{3}{2} kT}}$$

$$\lambda = \frac{h}{\sqrt{3m kT}}$$



3. For any particle carrying charge q accelerated by potential difference V volts

We know that

$$E = qV$$

Where

E = kinetic energy

de-broglie wavelength

$$\lambda = \frac{h}{\sqrt{2mE}}$$

$$\lambda = \frac{h}{\sqrt{2mqV}}$$

$$1 \text{ \AA} = 10^{-10} \text{ m}$$

4. De - Broglie wavelength of electron wave.

We know that

$$m_e = 9.1 \times 10^{-31} \text{ kg}$$

$$q = e = 1.6 \times 10^{-19} \text{ C}$$

de-broglie wavelength

$$\lambda = \frac{h}{\sqrt{2mqv}}$$

$$\lambda = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} \times V}}$$

$$\lambda = \frac{12.28}{\sqrt{V}} \text{ \AA}$$

Note : (i) Mass of Electron

$$m_e = 9.1 \times 10^{-31} \text{ kg}$$

(ii) Mass of proton

$$m_p = 1.67 \times 10^{-27} \text{ kg}$$

(iii) Mass of Neutron

$$m_n = 1.67 \times 10^{-27} \text{ kg}$$

(iv) Planck's constant

$$h = 6.63 \times 10^{-34} \text{ J-sec}$$

(v) Boltzmann's constant

$$k = 1.38 \times 10^{-23} \text{ J/K}$$

Q.1 Find the de-Broglie wavelength for an electron of energy V eV.

$$E = V \text{ eV}$$

$$E = V \times 1.6 \times 10^{-19} \text{ J}$$

$$E = 1.6 \times 10^{-19} V \text{ J}$$

We know that

$$\lambda = \frac{h}{\sqrt{2mE}}$$

$$\lambda = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} V}}$$

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

$$\lambda = \frac{12.28}{\sqrt{V}} \text{ \AA}$$

Q.2 Calculate the de-Broglie wavelength associated with an electron subjected to a potential difference of 50 volts.

Given

$$V = 50 \text{ V}$$

de-broglie wavelength

$$\lambda = \frac{12.28}{\sqrt{V}} \text{ \AA}$$

$$\lambda = \frac{12.28}{\sqrt{50}} \text{ \AA}$$

$$\lambda = 1.7366 \text{ \AA}$$

Q.3 Calculate wavelength of an α particle accelerated through a potential difference of 200 Volts.

Given

$$V = 200 \text{ V}$$

$$q = 2e = 2 \times 1.6 \times 10^{-19} \text{ C}$$

$$m = 4m_p = 4 \times 1.67 \times 10^{-27} \text{ kg}$$

de-broglie wavelength

$$\lambda = \frac{h}{\sqrt{2mqV}}$$

$$\lambda = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 4 \times 1.67 \times 10^{-27} \times 2 \times 1.6 \times 10^{-19} \times 200}}$$

$$\lambda = 0.00716 \text{ \AA}$$

$$1 \text{ \AA} = 10^{-10} \text{ m}$$

Q.4 Calculate the velocity and kinetic energy of a neutron having de-Broglie wavelength $1 \text{ \AA}$.

Given

$$\lambda = 1 \text{ \AA} = 10^{-10} \text{ m}$$

$$m_n = 1.67 \times 10^{-27} \text{ kg}$$

de-broglie wavelength

$$\lambda = \frac{h}{mv}$$

$$10^{-10} = \frac{6.63 \times 10^{-34}}{1.67 \times 10^{-27} \times v}$$

$$v = \frac{6.63 \times 10^{-34}}{10^{-10} \times 1.67 \times 10^{-27}}$$

$$v = 3.96 \times 10^3 \text{ m/sec}$$

Kinetic energy

$$E = \frac{1}{2} mv^2 = 1.67 \times 10^{-27} \times (3.96 \times 10^3)^2$$

$$E = 1.309 \times 10^{-20} \text{ J}$$

Q5 Calculate the de-Broglie wavelength associated with a proton moving with velocity equal to $\frac{1}{20}th$ velocity of light.

Given

$$v = \frac{1}{20} c = \frac{1}{20} \times 3 \times 10^8 \text{ m/s}$$

de-broglie wavelength

$$\lambda = \frac{h}{mv}$$

$$\lambda = \frac{h}{m_0 v \sqrt{1 - \frac{v^2}{c^2}}}$$

$$\lambda = \frac{h}{m_0 v} \sqrt{1 - \frac{v^2}{c^2}}$$

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\lambda = \frac{6.63 \times 10^{-34}}{1.67 \times 10^{-27} \times \frac{1}{20} \times 3 \times 10^8} \sqrt{1 - \left(\frac{1}{20}\right)^2}$$

$$\lambda = \frac{6.63 \times 10^{-34} \times 20}{1.67 \times 10^{-27} \times 3 \times 10^8} \sqrt{1 - \frac{1}{400}}$$

$$\lambda = 0.000264 \text{ \AA}$$

Q.6 Calculate the de-Broglie wavelength of neutron having kinetic energy, 1 eV.

Given

$$E = 1 \text{ eV} = 1.67 \times 10^{-19} \text{ J}$$

Rest mass Energy of neutron

$$E = mc^2$$

$$E = 1.67 \times 10^{-27} \times (3 \times 10^8)^2 \text{ J}$$

$$E = \frac{1.67 \times 10^{-27} \times (3 \times 10^8)^2}{1.6 \times 10^{-19}}$$

$$E = 939375000 \text{ eV}$$

So, ignored Relativistic concept

de-broglie wavelength

$$\lambda = \frac{h}{\sqrt{2mE}} = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 1.67 \times 10^{-27} \times 1.67 \times 10^{-19}}}$$

$$\lambda = 0.287 \text{ \AA}$$

Properties of Matter waves

$$\lambda = \frac{h}{mv}$$

- The de - Broglie wavelength of a wave associated with *Lighter particle* is greater than the wavelength associated with *heavier particle*.
- The de - Broglie wavelength of a wave associated with *slow moving particle* is greater than the wavelength associated with *fast moving particle*.
- If $v = 0$, then $\lambda = \infty$, Wave become indeterminate and if $v = \infty$, then $\lambda = 0$, This shows that the matter waves are generated only when the material particles are in motion.
- The matter waves are not electromagnetic in nature.
- The matter waves are generated by moving charged particles as well as by moving neutral particles.

<i>Matter wave</i>	<i>Electromagnetic wave</i>
(i) Matter waves are associated with moving particle and does not depend on charge (ii) Matter waves do not have any associated electric and magnetic field. (iii) Matter waves do not travel through the vacuum. (iv) The velocity of matter waves is generally less than the velocity of light. (v) Example – Electron beam	(i) Electromagnetic waves are associated with accelerated charge particle. (ii) Electromagnetic waves have electric and magnetic fields associated with them (iii) Electromagnetic waves travel through the vacuum. (iv) The velocity of electromagnetic waves is equal to the velocity of light (v) Example – Radio waves, UV light

Q.1 Calculate the de-Broglie wavelength associated with an electron subjected to a potential difference of 100 volts.

$$\lambda = 1.288 \text{ \AA}$$

Q.2 The kinetic energy of an electron is $4.55 \times 10^{-25} \text{ J}$ Calculate the velocity, momentum and wavelength of electron.

$$v = 1000 \text{ m/s}$$

$$p = 9.1 \times 10^{-28} \text{ kg m/s}$$

$$\lambda = 7.286 \times 10^{-7} \text{ m}$$

Q.3 A proton is moving with a velocity $2 \times 10^8 \text{ m/s}$. Calculate the wavelength of matter wave associated with it.

$$\lambda = 1.47 \times 10^{-5} \text{ \AA}$$

Q.4 Calculate the de-Broglie wavelength associated with nitrogen at 3.0 atmospheric pressure and 27° C mass of N_2 -atom = 4.65×10^{-26}

$$\lambda = 0.2754 \text{ \AA}$$

B.TECH : Engineering Physics

UNIT : Quantum Mechanics

Lecture-5

Today's target

- Davisson - Germer Experiment
- DPP
- PYQ

Gateway Classes

Davisson and Germer Experiment

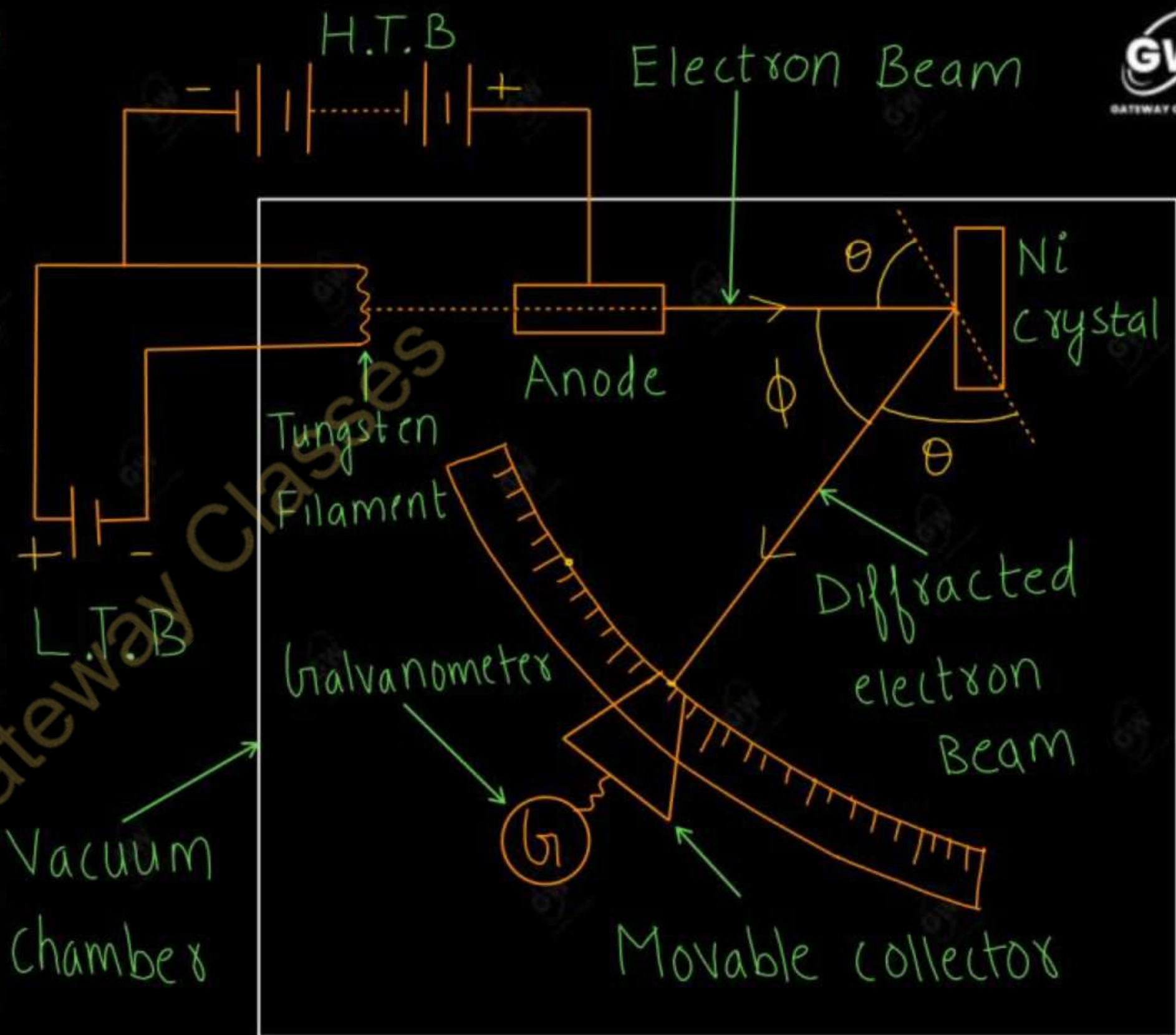
➤ In 1927, Davisson and Germer gave first experimental evidence of matter waves predicted by de-Broglie in 1924.

➤ They not only confirmed the existence of waves associated with fast moving electrons but also measured the wavelength of matter wave.

$$\lambda = \frac{h}{p}$$

➤ The experimental setup consist of an electron gun having tungsten filament **F** which is heated by passing electric current from low tension battery (L.T.B.)

➤ Due to thermionic emission electrons are emitted by the filament and these electrons are then accelerated under suitable potential difference provided by high tension battery (H.T.B.)



Set-up of Davisson-Germer Experiment

- The narrow beam of electrons comes out from cylinder is allowed to fall on the surface of a nickel crystal.
- The Electrons, scattered in different directions by the atoms of nickel crystal, are received by a movable detector which is just an electron collector.
- This electron collector is connected with a galvanometer (G).
- The deflection in G is directly proportional to intensity of electron beam entering the collector.
- The entire apparatus is enclosed in vacuum chamber.
- This experiment was performed at different accelerating voltage (V) but significant intensity of electron was observed around 44 V - 68 V.

Result of Davisson and Germer Experiment

Direction of
incident electron



44V



48V



54V



64V

Above Graphs show the intensity of electrons as a function of scattering angle ϕ

- (i) Intensity of scattered electron depends upon angle of scattering ϕ .
- (ii) A bump begins to appear in curve at 44 V.
- (iii) It was noticed that a strong peak appeared in the intensity (I) of the scattered electron for an accelerating voltage of 54 V at a scattering angle 50° .
- (iv) The size of bump start decreasing with further increase in accelerating voltage.
- (v) From the electron diffraction measurements, the wavelength of matter waves was found to be 0.165nm

Conclusion : The maximum intensity obtained in a particular direction shows the constructive interference.

So, for crystal constructive interference, Bragg's Law stated that

$$2d \sin \theta = n \lambda \quad \text{--- ①}$$

Where

d = interatomic separation
for Ni crystal

$$d = 0.914 \text{ \AA}$$

$$n = 1$$

From figure

$$\theta + \theta + \phi = 180$$

$$2\theta + 50 = 180$$

$$\theta = 65^\circ$$

Put these values in ①

$$2 \times 0.914 \text{ \AA} \times \sin 65 = \lambda$$

$$\lambda = 1.66 \text{ \AA}$$

By de-Broglie Hypothesis

$$\lambda = \frac{12.28}{\sqrt{V}}$$

$$\lambda = \frac{12.28}{\sqrt{54}}$$

$$\lambda = 1.67 \text{ \AA}$$

Thus, there is an excellent agreement between the theoretical value and the experimentally obtained value of De-broglie wavelength.

Q.1 What was the objective of conducting Davisson – Germer experiment ?

Q.2 Describe the experiment of Davisson and Germer to demonstrate the wave character of electrons.

Gateway Classes

B.TECH : Engineering Physics

UNIT : Quantum Mechanics

Lecture-6

Today's target

- Phase Velocity and Group Velocity
- DPP
- PYQ

Gateway Classes

Phase Velocity

- The velocity with which a monochromatic wave (single frequency and wavelength) travel through a medium is known as phase velocity or wave velocity.
- It is the characteristic of individual wave.
- **Expression for Phase Velocity**

Consider a wave whose displacement y is expressed as

$$y = a \sin(\omega t - kx)$$

Where

a = Amplitude

ω = Angular Frequency
 $= 2\pi\nu$

K = Propagation constant
 OR

Wave Vector

$$= \frac{2\pi}{\lambda}$$

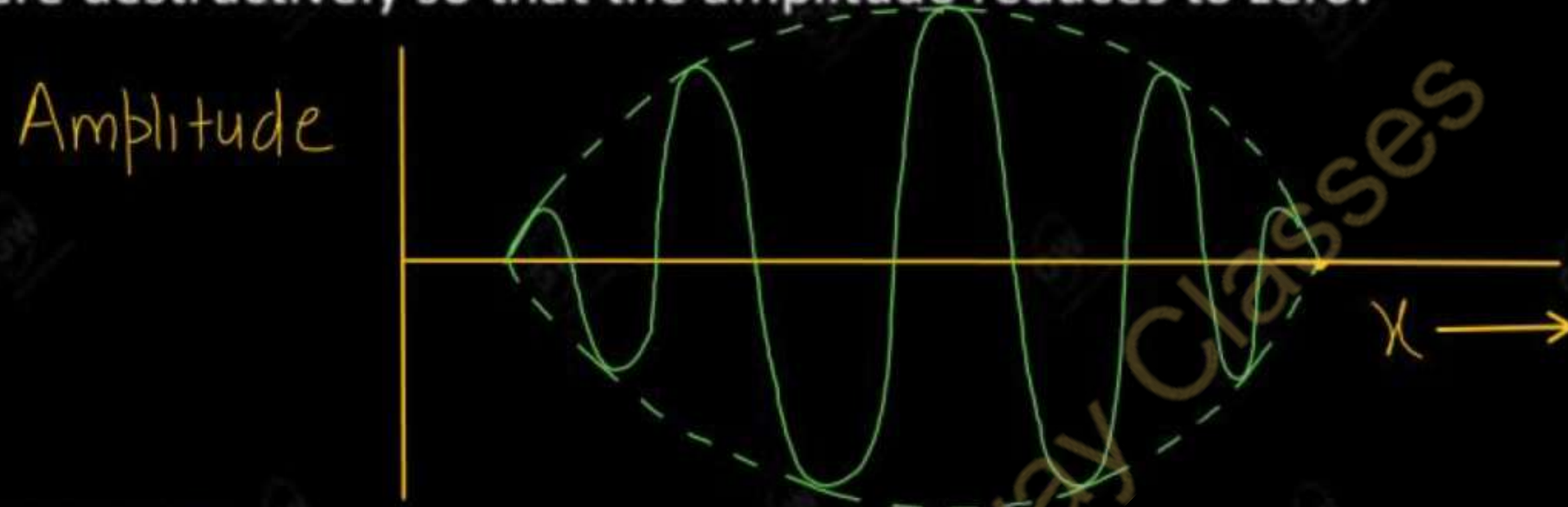
For the plane of constant phase

$$\omega t - kx = \text{constant}$$

diff. wrt t

A wave packet is a group of waves with slightly different wavelength and velocity such that they interfere

Constructively over a small region of space where the particle can be located and outside this space they interfere destructively so that the amplitude reduces to zero.

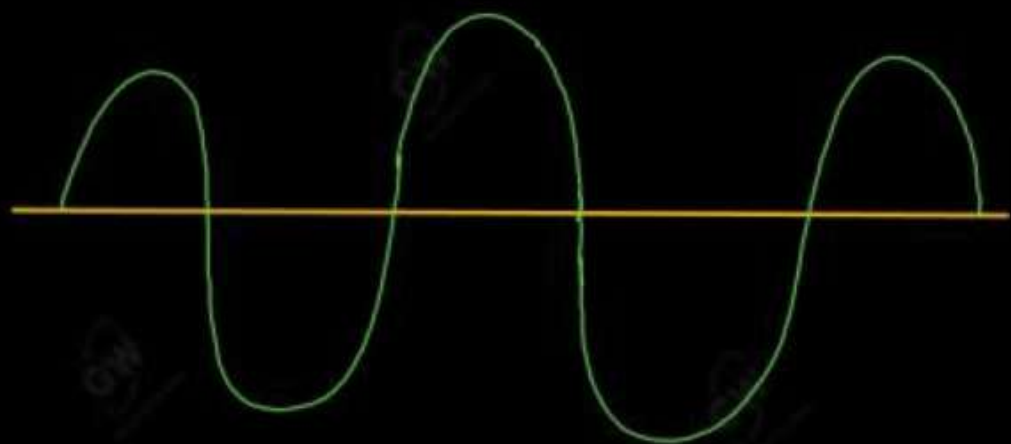


Group Velocity

- The velocity with which a wave packet (group of waves) travel in a medium is known as group velocity.
- It is the characteristic of group of waves.

$$v_g = \frac{d\omega}{dk}$$

Let us Consider two waves whose having same amplitude and different angular frequency or wavelength

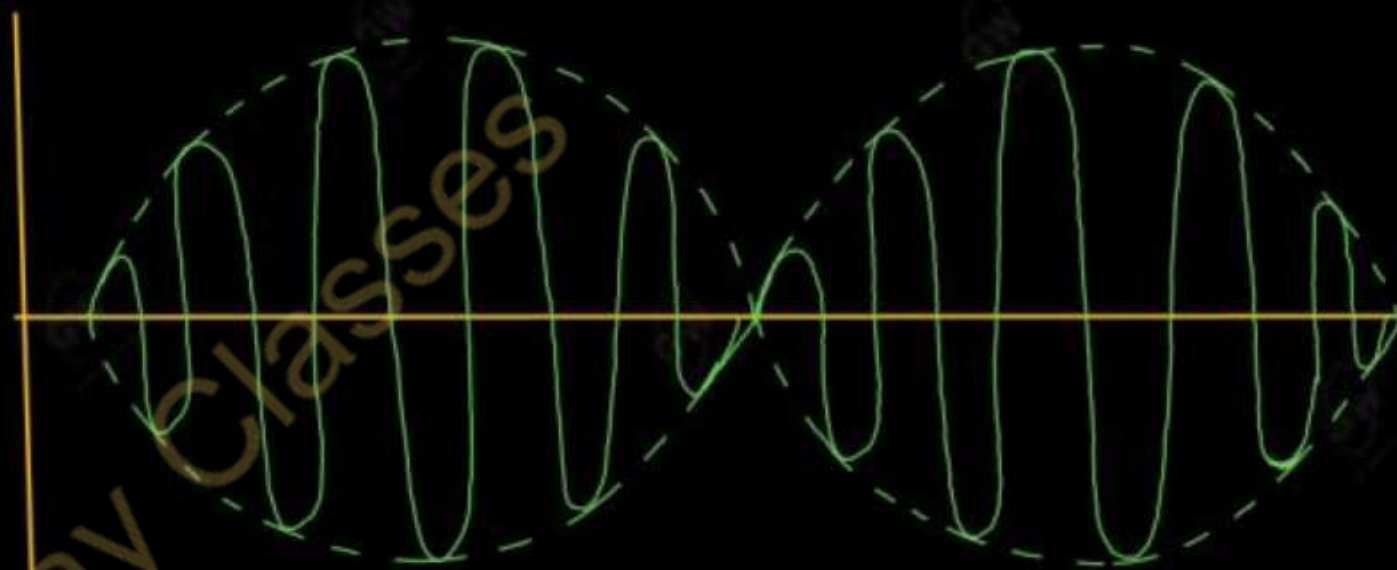


$$y_1 = a \sin(\omega_1 t - k_1 x)$$



$$y_2 = a \sin(\omega_2 t - k_2 x)$$

Apply principle of superposition



$$y_1 + y_2 = a \sin(\omega_1 t - k_1 x) + a \sin(\omega_2 t - k_2 x)$$

$$y_1 + y_2 = a [\sin(\omega_1 t - k_1 x) + \sin(\omega_2 t - k_2 x)]$$

$$\therefore \sin C + \sin D = 2 \sin\left(\frac{C+D}{2}\right) \cos\left(\frac{C-D}{2}\right)$$

$$y_1 + y_2 = a \left[2 \sin\left(\frac{\omega_1 t - k_1 x + \omega_2 t - k_2 x}{2}\right) \cos\left(\frac{\omega_1 t - k_1 x - \omega_2 t + k_2 x}{2}\right) \right]$$

$$y_1 + y_2 = 2a \left[\sin\left\{\left(\frac{\omega_1 + \omega_2}{2}\right)t - \left(\frac{k_1 + k_2}{2}\right)x\right\} \cos\left\{\left(\frac{\omega_1 - \omega_2}{2}\right)t - \left(\frac{k_1 - k_2}{2}\right)x\right\} \right]$$

$$y_1 + y_2 = y, \quad \frac{\omega_1 + \omega_2}{2} = \omega, \quad \frac{k_1 + k_2}{2} = K, \quad \frac{\omega_1 - \omega_2}{2} = \frac{d\omega}{2}, \quad \frac{k_1 - k_2}{2} = \frac{dK}{2}$$

$$y = 2a \sin(\omega t - Kx) \cos\left(\frac{d\omega}{2}t - \frac{dK}{2}x\right)$$

$$y = 2a \cos\left(\frac{d\omega}{2}t - \frac{dK}{2}x\right) \sin(\omega t - Kx)$$

Where

$$A = 2a \cos\left(\frac{d\omega}{2}t - \frac{dK}{2}x\right)$$

$$y = A \sin(\omega t - Kx)$$

Group velocity (V_g)

$$V_g = \frac{\frac{d\omega}{2}}{\frac{dK}{2}}$$

$$V_g = \frac{d\omega}{dK}$$

Gateway Classes

Relation between Phase Velocity and Group Velocity

We know that
group velocity (V_g)

$$V_g = \frac{d\omega}{dk} \quad \text{--- (1)}$$

Phase velocity

$$V_p = \frac{\omega}{k}$$

$$\omega = k V_p$$

diff wrt k

$$\frac{d\omega}{dk} = k \frac{dV_p}{dk} + V_p$$

Put $\frac{d\omega}{dk}$ in (1)

$$V_g = V_p + k \frac{dV_p}{dk} \quad \text{--- (2)}$$

We know that

$$k = \frac{2\pi}{\lambda}$$

diff wrt λ

$$\frac{dk}{d\lambda} = -\frac{2\pi}{\lambda^2}$$

$$dk = -\frac{2\pi}{\lambda^2} d\lambda$$

Put k and dk in (2)

$$V_g = V_p + \frac{2\pi}{\lambda} \times \frac{dV_p}{\left(-\frac{2\pi}{\lambda^2}\right) d\lambda}$$

$$V_g = V_p - \lambda \frac{dV_p}{d\lambda}$$

↑
Required Relation

For non - dispersive medium

$$dV_p = 0$$

$$V_g = V_p - 0$$

$$V_g = V_p$$

Q.1 What do you mean by group velocity?

Q.2 Define wave packet with neat diagram.

Q.3 Distinguish between group velocity and phase velocity. Establish a relation between them in a dispersive medium. What will be relation between these velocities in non – dispersive medium?

B.TECH : Engineering Physics

UNIT : Quantum Mechanics

Lecture-7

Today's target

➤ Schrodinger wave equation

(i) Time In-dependent Schrodinger wave equation

(ii) Time Dependent Schrodinger wave equation

➤ *Born interpretation of wave function*

➤ DPP

➤ PYQ

Schrodinger wave equation

➤ Wave function

The state of the particle at any instant is represented by a wave function (Ψ)

OR

The quantity whose variation builds up matter waves is called wave function (Ψ)

OR

The wave function (Ψ) describe the behaviour of wave associated with moving particle

- Schrodinger wave equations describe the wave motion of matter wave associated with particle and also known as Equation of Motion of Matter Waves.
 - (i) Time In-dependent Schrodinger wave equation
 - (ii) Time Dependent Schrodinger wave equation

ψ si

(i) Time dependent Schrodinger wave equation

The differential equation of a wave motion of a Particle in one dimension can be written as

$$\frac{\partial^2 \psi}{\partial x^2} = -\frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2} \quad \text{--- (1)}$$

The general solution of (1)

$$\psi(x, t) = A e^{-i\omega(t - \frac{x}{v})}$$

Put

$$\omega = 2\pi\nu$$

$$v = \nu\lambda$$

$$\psi(x, t) = A e^{-i2\pi\nu(t - \frac{x}{v\lambda})}$$

$$\psi(x, t) = A e^{-2\pi i(\nu t - \frac{x}{\lambda})}$$

Put $\nu = \frac{E}{h}$ $\left\{ \because E = h\nu \right\}$

$$\lambda = \frac{h}{p}$$

$$\psi(x, t) = A e^{-2\pi i\left(\frac{E}{h}t - \frac{px}{h}\right)}$$

$$\psi(x, t) = A e^{-\frac{2\pi i}{h}(Et - px)}$$

$$\psi(n, t) = A e^{-\frac{i}{\hbar/2\pi} (Et - pn)}$$

Put $\frac{h}{2\pi} = \hbar$

$$\psi(n, t) = A e^{-\frac{i}{\hbar} (Et - pn)}$$

— (2)

Diff. wrt n

$$\frac{\partial \psi}{\partial n} = A e^{-\frac{i}{\hbar} (Et - pn)} \times \left(\frac{i}{\hbar} p \right)$$

Again diff. wrt n

$$\frac{\partial^2 \psi}{\partial n^2} = A \left(\frac{i}{\hbar} p \right) e^{-\frac{i}{\hbar} (Et - pn)} \times \left(\frac{i}{\hbar} p \right)$$

$$\frac{\partial^2 \psi}{\partial n^2} = -\frac{A p^2}{\hbar^2} e^{-\frac{i}{\hbar} (Et - pn)}$$

$$\frac{\partial^2 \psi}{\partial n^2} = -\frac{p^2}{\hbar^2} A e^{-\frac{i}{\hbar} (Et - pn)}$$

$$\frac{\partial^2 \psi}{\partial n^2} = -\frac{p^2 \psi}{\hbar^2} \quad (\text{From (2)})$$

$$p^2 = -\frac{\hbar^2}{\psi} \frac{\partial^2 \psi}{\partial n^2} \quad (3)$$

diff eqn (2) wrt t

$$\frac{\partial \psi}{\partial t} = A e^{\frac{-i}{\hbar}(Et - px)} \times \left(-\frac{i}{\hbar} E\right)$$

$$\frac{\partial \psi}{\partial t} = \psi \left(-\frac{i}{\hbar} E\right) \quad \text{From (2)}$$

$$E = \frac{-\hbar}{\psi i} \frac{\partial \psi}{\partial t}$$

$$E = \frac{\hbar i}{\psi} \frac{\partial \psi}{\partial t} \quad \text{--- (4)}$$

Apply conservation of Energy

$$E = K.E + P.E$$

$$E = \frac{p^2}{2m} + V \quad \text{--- (5)}$$

Put p^2 and E in (5)

$$\frac{\hbar i}{\psi} \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m\psi} \frac{\partial^2 \psi}{\partial x^2} + V$$

Multiply both side by ψ

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi$$

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi = i\hbar \frac{\partial \psi}{\partial t}$$

Required Equation

In 3-Dimension

$$-\frac{\hbar^2}{2m} \left(\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} \right) + V\psi = i\hbar \frac{\partial \psi}{\partial t}$$

$$-\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi = i\hbar \frac{\partial \psi}{\partial t}$$

$$\left(-\frac{\hbar^2}{2m} \nabla^2 + V \right) \psi = \left(i\hbar \frac{\partial}{\partial t} \right) \psi$$

Where

$-\frac{\hbar^2}{2m} \nabla^2 + V \rightarrow$ Hamiltonian operator

$i\hbar \frac{\partial}{\partial t} \rightarrow$ Energy operator

(ii) Time In-dependent Schrodinger wave equation

The differential equation of a wave motion of a Particle in one dimension can be written as

$$\frac{\partial^2 \psi}{\partial x^2} = -\frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2} \quad \text{--- ①}$$

The General Solution of eqⁿ ①

$$\psi(x, t) = A e^{-i\omega(t - \frac{x}{v})}$$

$$\begin{aligned} \text{Put } \omega &= 2\pi\nu \\ v &= \nu\lambda \end{aligned}$$

$$\psi(x, t) = A e^{-i2\pi\nu(t - \frac{x}{v\lambda})}$$

$$\psi(x, t) = A e^{-2\pi i \left(\nu t - \frac{x}{\lambda} \right)}$$

$$\text{Put } \nu = \frac{E}{h} \quad \left\{ \because E = h\nu \right\}$$

$$\lambda = \frac{h}{p}$$

$$\psi(x, t) = A e^{-2\pi i \left(\frac{Et}{h} - \frac{px}{h} \right)}$$

$$\psi(x, t) = A e^{-\frac{2\pi i}{h} (Et - px)}$$

$$\psi(x, t) = A e^{-\frac{i}{h/2\pi} (Et - px)}$$

Put $\frac{h}{2\pi} = \hbar$

$$\psi(x, t) = A e^{-\frac{i}{\hbar}(Et - px)}$$

②

Diff. wrt x

$$\frac{\partial \psi}{\partial x} = A e^{-\frac{i}{\hbar}(Et - px)} \times \frac{i}{\hbar} p$$

Again diff. wrt x

$$\frac{\partial^2 \psi}{\partial x^2} = A e^{-\frac{i}{\hbar}(Et - px)} \times \left(\frac{i}{\hbar} p\right)^2$$

$$\frac{\partial^2 \psi}{\partial x^2} = -\frac{p^2}{\hbar^2} A e^{-\frac{i}{\hbar}(Et - px)}$$

$$\frac{\partial^2 \psi}{\partial x^2} = -\frac{p^2}{\hbar^2} \psi \quad (\text{From eqn ②})$$

$$p^2 = -\frac{\hbar^2}{\psi} \frac{\partial^2 \psi}{\partial x^2} \quad \text{--- ③}$$

Apply Principle of conservation of energy

$$E = K.E + P.E$$

$$E = \frac{p^2}{2m} + V$$

Put p^2 in above equation

$$E = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V$$

$$E\psi = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi$$

$$\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + (E - V)\psi = 0$$

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{2m}{\hbar^2} (E - V)\psi = 0$$

Required equation

In 3-D

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} + \frac{2m}{\hbar^2} (E - V)\psi = 0$$

$$\nabla^2 \psi + \frac{2m}{\hbar^2} (E - V)\psi = 0$$

For Free particle

$$V = 0$$

$$\nabla^2 \psi + \frac{2m}{\hbar^2} E\psi = 0$$

Physical interpretation of wave function or Born interpretation of wave function

➤ The state of the particle at any instant is represented by a wave function

OR

The quantity whose variation builds up matter waves is called wave function.

- The wave function is a complex function and the satisfactory interpretation of the wave function ' Ψ ' was given by Max Born in 1926 .
- According to him the probability of finding a particle at any point in space (x, y, z) at a given time t is proportional to $|\Psi|^2$, which is called probability density and ' Ψ ' is probability amplitude.
- Since the particle is certainly somewhere in space, the total probability of finding the particle in space is unity, i. e.

$$\iiint |\Psi|^2 dx dy dz = 1$$

➤ This is called normalization condition and ' Ψ ' is said to be normalized .

- The value of energy of which Schrodinger equation (time independent) can be solved are called 'eigen values' and the corresponding wave function are called "eigen function".
- Characteristic of wave function.
 - (i) It must be normalized.
 - (ii) It must be finite everywhere.
 - (iii) It must be single – valued.
 - (iv) It must be continuous and its first derivative should also be continuous.

Q.1 Derive time – dependent schrodinger's wave equation.

Q.2 Derive time – independent schrodinger's wave equation.

Q.3 Explain the physical significance of wave function given by Max Born.

Gateway Classes

B.TECH : Engineering Physics

UNIT : Quantum Mechanics

Lecture-8

Today's target

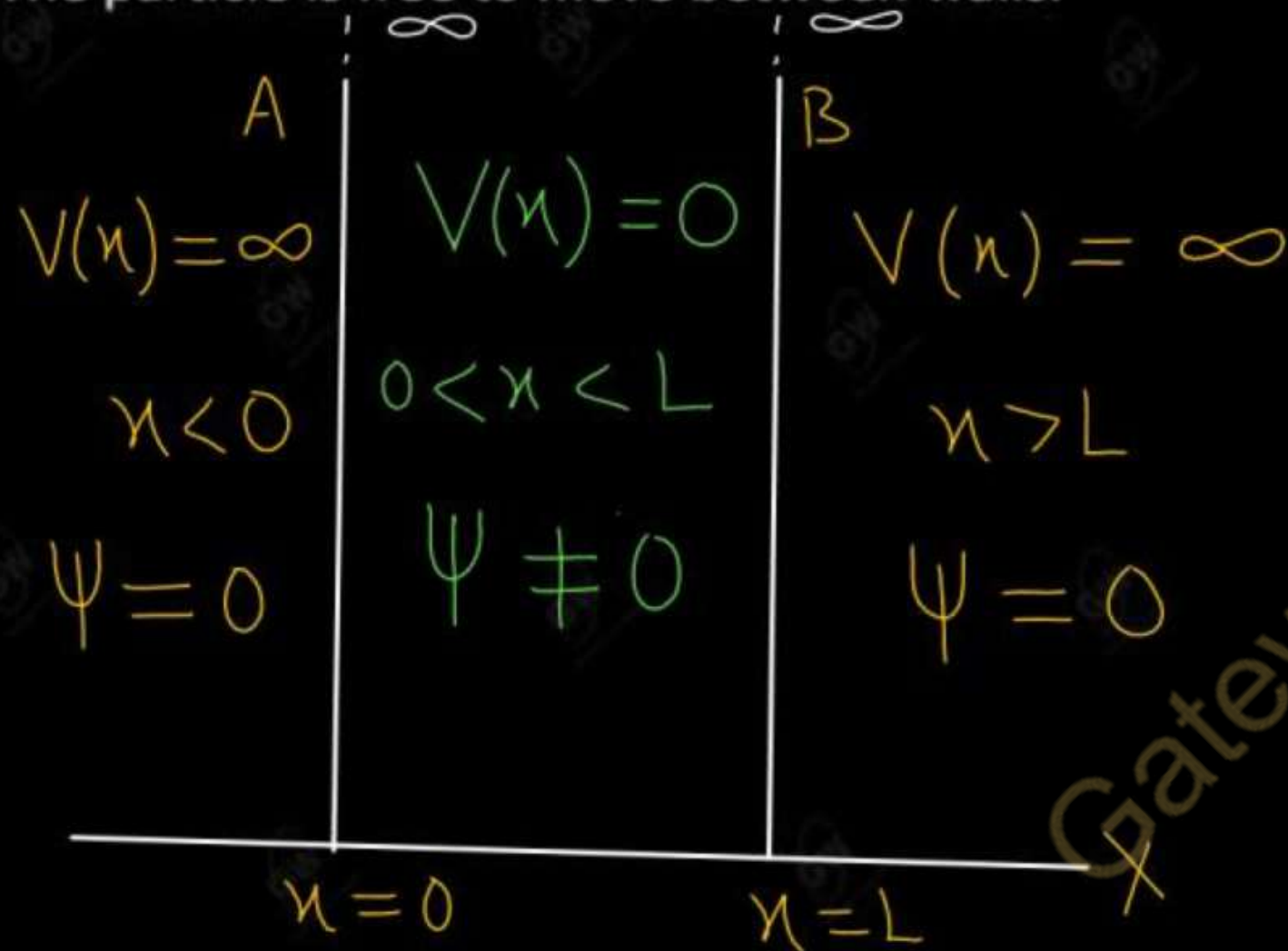
- Particle in one dimensional box
- DPP
- PYQ

Gateway Classes

Particle in one dimensional box (Application of schrodinger wave equation)

Let us consider a particle of mass 'm' moving along x-axis between two rigid walls A and B at $x = 0$ and $x = L$.

The particle is free to move between walls.



Potential Function

$$V(x) = \begin{cases} 0 & 0 < x < L \\ \infty & x < 0 \text{ and } x > L \end{cases}$$

Using Schrodinger time independent
Wave equation for one-dimension

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{2m}{\hbar^2} (E - V) \psi = 0$$

OR

$$\frac{d^2 \psi}{dx^2} + \frac{2m}{\hbar^2} (E - V) \psi = 0$$

For Free particle ($V=0$)

$$\frac{d^2 \psi}{dx^2} + \frac{2m}{\hbar^2} E \psi = 0$$

Put $\frac{2mE}{\hbar^2} = k^2$ — (1)

$$\frac{d^2 \psi}{dx^2} + k^2 \psi = 0 \quad \text{--- (2)}$$

General solution of eqⁿ (2)

$$\psi = A \sin kx + B \cos kx \quad \text{--- (3)}$$

Apply First Boundary condition

At $x=0, \psi=0$

$$0 = B + 0$$

$$B = 0$$

Put $B = 0$ in (3)

$$\psi = A \sin kx \quad \text{--- (4) Eigen Functions}$$

Apply second Boundary condition

$$\text{At } x=L, \psi=0$$

$$0 = A \sin KL + 0$$

$$A \sin KL = 0$$

$$\therefore A \neq 0$$

$$\therefore \sin KL = 0$$

$$\Rightarrow KL = \pm n\pi$$

$$\boxed{K = \pm \frac{n\pi}{L}} \quad \text{--- (5)}$$

Put K in (1)

$$\frac{2mE}{\hbar^2} = \left(\frac{n\pi}{L}\right)^2$$

$$\frac{2mE}{\left(\frac{h}{2\pi}\right)^2} = \frac{n^2 \pi^2}{L^2}$$

$$\frac{2mE \times 4\pi^2}{h^2} = \frac{n^2 \pi^2}{L^2}$$

$$\boxed{E_n = \frac{n^2 h^2}{8mL^2}}$$

Where

$$n = 1, 2, 3, \dots$$

Hence

Energy of the particle is in discrete form

$$E_1 = \frac{h^2}{8mL^2}$$

$$E_2 = \frac{4h^2}{8mL^2}$$

$$E_3 = \frac{9h^2}{8mL^2}$$

and so on

Here

$$E_1, E_2, E_3, \dots$$

are known as eigen values

Put k in (4)

$$\psi = A \sin \frac{n\pi x}{L} \quad \text{--- (6)}$$

(Eigen Function)

Using Normalisation

$$\int_0^L |\psi|^2 dx = 1$$

$$\int_0^L A^2 \frac{\sin^2 n\pi x}{L} dx = 1$$

$$\frac{A^2}{2} \int_0^L \left(1 - \cos \frac{2n\pi x}{L}\right) dx = 1 \quad \left\{ \sin^2 \theta = \frac{1 - \cos 2\theta}{2} \right\}$$

$$\frac{A^2}{2} \left(x - \frac{\sin \frac{2n\pi x}{L}}{\frac{2n\pi}{L}} \right)_0^L = 1$$

$$\frac{A^2}{2} [L - 0 - (0 - 0)] = 1$$

$$\frac{A^2 L}{2} = 1$$

$$A^2 L = 2$$

$$A^2 = \frac{2}{L}$$

$$A = \sqrt{\frac{2}{L}}$$

Put A in (6)

$$\psi = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}$$



Normalised wave function

OR

Eigen function

Graph of energy Level, Wave function (Ψ) and probability density ($|\Psi|^2$)

We know that

$$E_n = \frac{n^2 h^2}{8mL^2}$$

$$E_1 = \frac{h^2}{8mL^2}$$

$$E_2 = \frac{4h^2}{8mL^2} = 4E_1$$

$$E_3 = \frac{9h^2}{8mL^2} = 9E_1$$

$$E_4 = \frac{16h^2}{8mL^2} = 16E_1$$

$$E_5 = \frac{25h^2}{8mL^2} = 25E_1$$

$$E_2 - E_1 = 3E_1$$

$$E_3 - E_2 = 5E_1$$

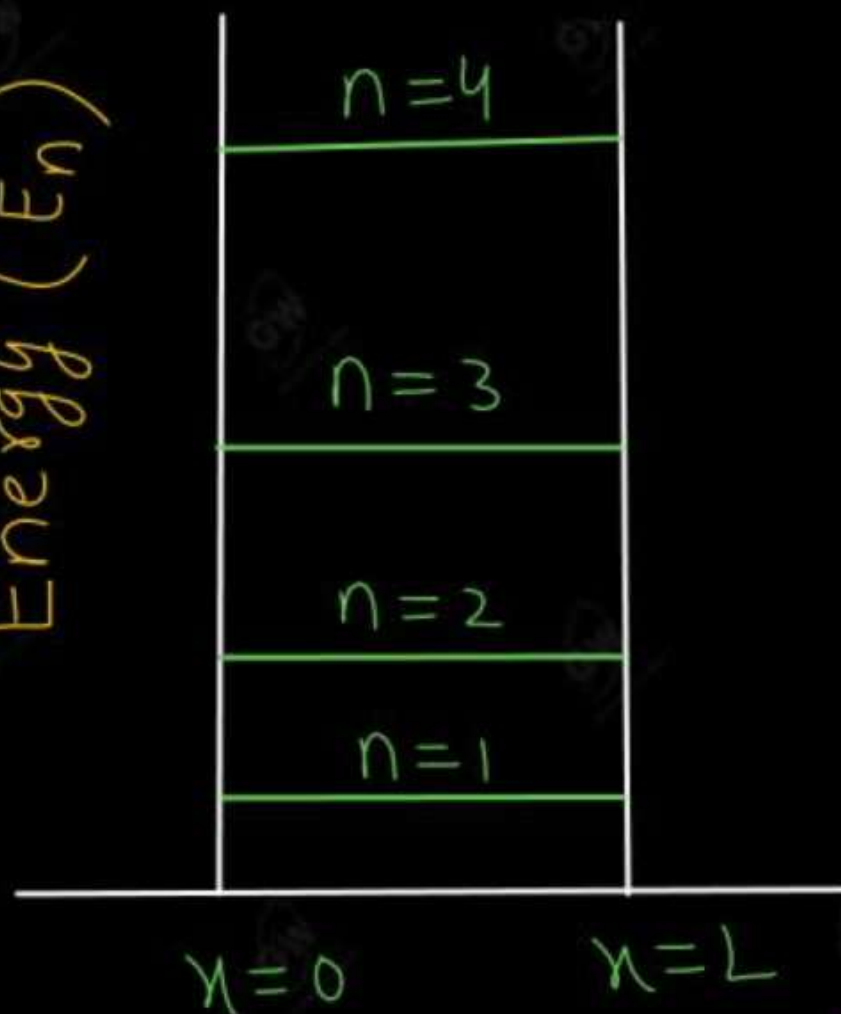
$$E_4 - E_3 = 7E_1$$

$$E_5 - E_4 = 9E_1$$

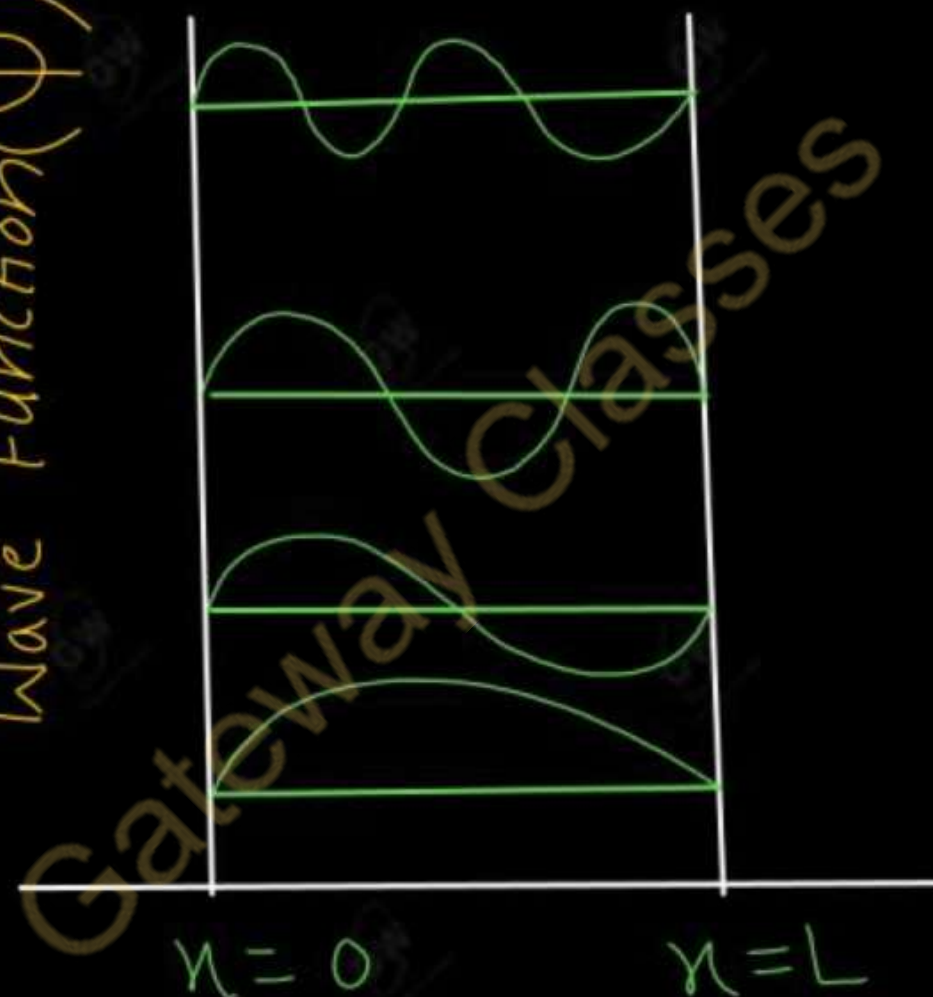
and so on

It is clear that the difference between energy of consecutive state is not constant. Therefore it is concluded that the energy level are not equally spaced.

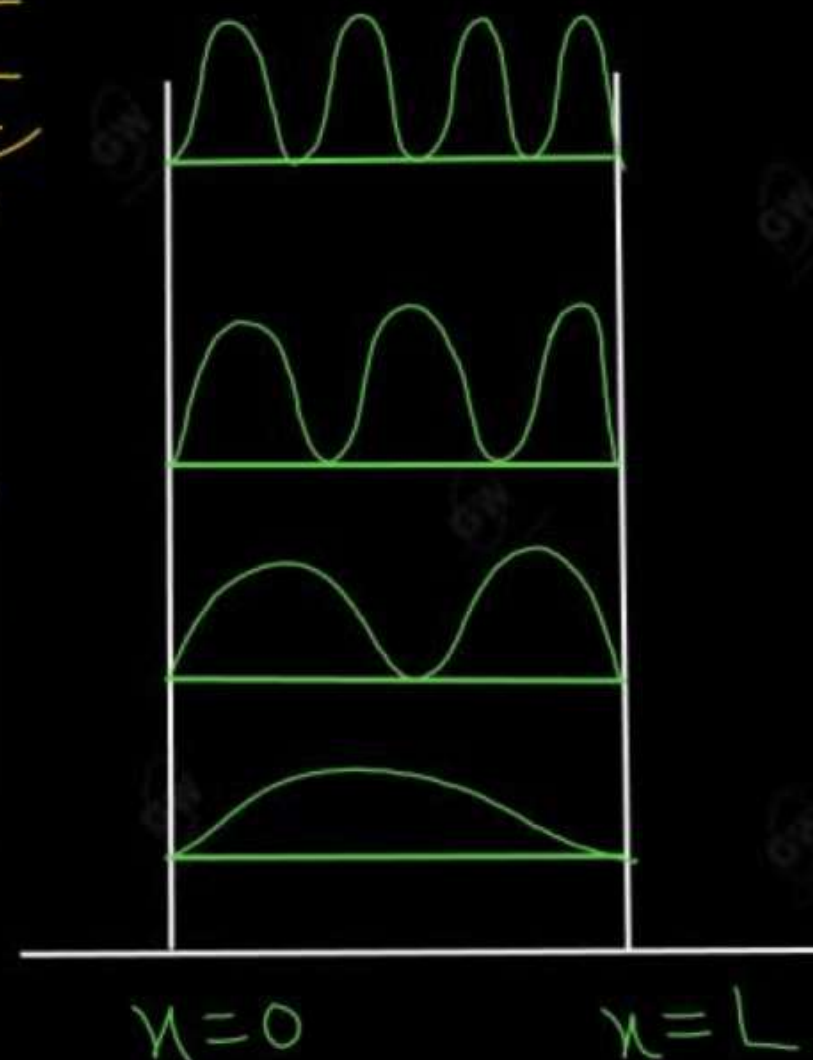
Energy (E_n)



Wave Function (ψ)



Probability Density ($|\psi|^2$)



$$\psi = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}$$

Q.1 An electron is bound in one dimensional potential box which has width $2.5 \times 10^{-10} \text{ m}$. Assuming the height of the box to be infinite, calculate the lowest two permitted energy values of the electron.

Given

$$L = 2.5 \times 10^{-10} \text{ m}$$

$$m = 9.1 \times 10^{-31} \text{ kg}$$

$$h = 6.63 \times 10^{-34} \text{ J-sec}$$

We know that

$$E_n = \frac{n^2 h^2}{8mL^2}$$

$$E_n = \frac{n^2 \times (6.63 \times 10^{-34})^2}{8 \times 9.1 \times 10^{-31} \times (2.5 \times 10^{-10})^2}$$

$$E_n = \frac{0.0966 \times 10^{-17} n^2 \text{ J}}{1.6 \times 10^{-19}}$$

$$E_n = 6.04 n^2 \text{ eV}$$

$$E_1 = 6.04 \text{ eV}$$

$$E_2 = 24.15 \text{ eV}$$

Note : (i) Mass of Electron

$$m_e = 9.1 \times 10^{-31} \text{ kg}$$

(ii) Mass of proton

$$m_p = 1.67 \times 10^{-27} \text{ kg}$$

(iii) Mass of Neutron

$$m_n = 1.67 \times 10^{-27} \text{ kg}$$

(iv) Planck's constant

$$h = 6.63 \times 10^{-34} \text{ J-sec}$$

Q.2 What is the lowest energy that a neutron can have if confined to move along the edge of an impenetrable box of length 10^{-14} m.

Given

$$L = 10^{-14} \text{ m}$$

$$m = 1.67 \times 10^{-27} \text{ kg}$$

$$h = 6.63 \times 10^{-34} \text{ J-sec}$$

We know that

$$E_n = \frac{n^2 h^2}{8mL^2}$$

For Lowest energy

$$n = 1$$

$$E_1 = \frac{(1)^2 \times (6.63 \times 10^{-34})^2}{8 \times 1.67 \times 10^{-27} \times (10^{-14})^2}$$

$$E_1 = \frac{3.28 \times 10^{-13} \text{ J}}{1.6 \times 10^{-19}}$$

$$E_1 = 2.05 \text{ MeV}$$

Quantum Mechanics : Particle in one dimensional box

Q.1 Write down the time independent schrodinger wave equation for a particle in one – dimensional box (infinitely deep potential well) and find out energy eigen values (energy level) and corresponding energy eigen functions (normalized wave functions) of the particle.

Q.2 Find the two lowest permissible energy states for an electron which is confined in a one dimensional infinite potential box width $3.5 \times 10^{-9} \text{ m}$

$$E_1 = 3.08 \times 10^{-2} \text{ eV}$$

$$E_2 = 12.32 \times 10^{-2} \text{ eV}$$

Q.3 An electron is trapped in one dimensional region of length $1\text{\AA}$. Find the amount of energy that must be supplied to excite the electron from ground state to first excited state.

$$\Delta E = E_2 - E_1 = 113.1 \text{ eV}$$

Q.4 Calculate the energy difference between the ground state and the first excited state for an electron in a one-dimensional rigid box of length 10^{-8} cm .

$$\Delta E = E_2 - E_1 = 113.1 \text{ eV}$$

Session : 2024-25

Marks-2

Q.1 State the assumptions made by the Planck's to explain black body radiation curve.

Q.2 Describe the term wave packet in Quantum Mechanics.

Marks-7

Q.1 Derive Schrodinger's time independent wave equation for a free particle and give the physical interpretation of wave function.

Q.2 Describe the experiment of Davisson and Germer to demonstrate the wave character of electrons.

Q.3 A particle is moving in one dimensional potential box (of infinite height) of width $20 \text{ \AA}$. Calculate the probability of finding the particle within an interval of $10 \text{ \AA}$ at the center of the box when it is in its state of least energy.

Marks-2

Q.1 What is the physical significance of wave function?

Q.2 Explain basic postulates of Plank's law of radiation.

Marks-7

Q.1 Calculate the energy of an oscillator of frequency 5.3×10^{12} /sec. at 500 K treating it as (i) classical oscillator (ii) Planck's Oscillator. Given that $k = k_B = 1.38 \times 10^{-23}$ J/K and $h = h = 6.6 \times 10^{-34}$ J-Sec.

Q.2 Discuss Compton Effect & Derive an expression for Compton shift & show that Why Compton effect not observed with visible light.

Q.3 Find Eigen value & Eigen function of free particle in a box by using Schrödinger wave equation

Q.4 Calculate the Compton shift and kinetic energy of recoil electron if X-rays of wavelength $1.0 \text{ \AA}$ are scattered from a carbon block. The scattered radiation is viewed at 90° to the incident beam.

Q.5 Distinguish between phase velocity and group velocity of a wave packet and establish the relation between them.

Q.6 Find an expression for the energy state eigen value and wave function of a particle in one dimensional box.

2022-23

Marks-2

Q.1 Can Compton effect be observed with visible light? Explain briefly.

Q.2 What was the objective of conducting Davisson-Germer experiment?

Q.3 Write down the Planck's expression (formula) for spectral energy density in Black Body radiation.

Q.2 What do you understand by phase velocity and group velocity of waves ?

Marks-7

Q.1 Discuss any three physical phenomenon which could not be explained on the basis of classical physics.

Q.2 Distinguish between group velocity and phase velocity. Establish a relation between them in a dispersive medium. What will be relation between these velocities in non dispersive medium?

- Q.3** Explain the physical significance of wave function given by Max Born.
- Q.4** Solve Schrödinger equation for a free particle in one dimensional box and show that the energy Eigen values are discrete. An electron is trapped in one dimensional region of length $1.0 \text{ \AA}$. Find the amount of energy that must be supplied to excite the electron from ground state to first excited state.
- Q.5** Derive an expression for Compton wavelength shift ($\Delta\lambda$) for a Compton scattering experiment. The wavelength of an X-ray photon is doubled on being scattered through 90° with a carbon block in a Compton Experiment. Find out the wavelength of the incident photon. (Electron mass $m_e = 9.1 \times 10^{-31} \text{ kg}$, Planck's constant $h = 6.63 \times 10^{-34} \text{ J s}$, speed of light $c = 3.0 \times 10^8 \text{ m/s}$).
- Q.6** Derive time independent Schrodinger wave equation. Write down the time independent Schrodinger equation for a particle in one-dimensional box (infinitely deep potential well) and find out energy eigenvalues (energy levels) and the corresponding energy Eigen functions (normalized wave functions) of the particle.

Subjects By Gulshan sir in App

S.NO	SUBJECT	SEMESTER
1	Engineering Mathematics – I	I
2	Engineering Mathematics – II	II
3	Engineering Physics	I/II
4	Fundamental of electronics engineering	I/II
5	Engineering Mathematics – III	III / IV
6	Engineering Mathematics – IV	III / IV
7	Discrete Mathematics OR Discrete Structures and Theory of Logic (DSTL)	III



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